

Jatin Bhanot

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PROFILE SUMMARY

Computer Science undergraduate with a systematic focus on machine learning pipeline design and hardware-constrained inference. Possesses foundational agility across C++, Python, and machine Learning—applied across embedded inference stacks, distributed backends, and controlled ML experimentation. Research interests span edge-optimised vision systems, spatio-temporal traffic analytics, and retrieval-augmented generation, pursued across active international research environments in Taiwan and Kazakhstan.

EDUCATION

Chitkara University B.E. Computer Science (GPA: 3.46/4.0)	Chandigarh, India 2022 – 2026
National Formosa University Exchange & FTIP (Edge AI & UAV Vision)	Taiwan Feb – Jun 2026
Astana IT University Exchange & Research Internship (Applied CS)	Kazakhstan Mar – Jun 2025

EXPERIENCE

Research Intern – Edge AI & UAV Vision | National Formosa University (Formosa TIP), Taiwan Feb 2026 – Jun 2026

- Leading system integration across AI, data, and hardware modules for a UAV-based urban traffic analytics platform; coordinating multi-module team execution toward a deployable research prototype.
- End-to-end pipeline: VisDrone-trained YOLO11n (5 classes, conf=0.22) + ByteTrack over 2,049 frames — 113K detections, 2,527 unique vehicle IDs, avg 55veh/frame; SAHI for small-object aerial inference.
- Congestion classifier (96.6% SLOW-3.2% CONGESTED) + parking detector (174 checked, 15 flagged, 0 false positives); Engineered a custom bounding-box heuristic to filter and classify unmapped vehicle categories (bicycles) without retraining.

PROJECTS

Tomato Disease Classification via Class-Aware Selective Mixing | Independent Research Feb 2026 – Ongoing

- Controlled 5-strategy ablation on MobileNetV2 transfer learning (PlantVillage, 18,160 images, 10 classes) to isolate the effect of input representation on classification performance.
- Proposed Class-Aware Selective Mixing routing each class to its empirically optimal visual domain — **93.88% accuracy**, F1 macro 0.940, +4.82% over colour-only baseline, *Paper in progress*.

Microservices E-Commerce Backend | Golang, gRPC, PostgreSQL

- Engineered a modular microservices backend in Golang using gRPC to empirically evaluate inter-service communication overhead, database design trade-offs, and concurrent scalability under clean architecture constraints.

Carbon Credit Marketplace | Web3, Smart Contracts

- Designed a decentralised system to analyse transparency and auditability in carbon credit transactions; evaluated data immutability vs. system efficiency trade-offs in smart contract architecture. This project helped me to secure top 10 position in HackIndia Spark 5.

RESEARCH INTERESTS

- Sparse Personalized AI (NeuroPACA Architecture):** Applying Lottery Ticket Hypothesis pruning and behavioral usage graphs to extract ultra-lightweight, zero-latency LLM subnetworks for always-on edge co-pilots.
- Hardware-Constrained Edge Vision:** Optimizing low-latency detection pipelines (YOLO, EfficientDet-lite, MobileNet) for micro-scale embedded deployment and severe aerial occlusion.
- Applied LLM Infrastructure Backends:** Architecting precision Retrieval-Augmented Generation (RAG) pipelines and modular microservices (gRPC, PostgreSQL) under clean architecture constraints.

SKILLS

Languages: C/C++, Python, Rust, Golang **AI/ML:** PyTorch, LLaMA, RAG, Vector DBs, YOLO, MobileNet

Systems: gRPC, RESTful APIs, Microservices, PostgreSQL, MongoDB, Smart Contracts **Tools:** Git, Linux

ACHIEVEMENTS & ACTIVITIES

- HackIndia Spark 5:** Top-10 finish among 100+ teams; recognised for system design depth and implementation under competition constraints.
- IEEE SIST 2025 — Volunteer:** Organised sessions at an international technical conference; supported academic networking and logistics.
- NSS Club:** Community outreach and collaborative initiatives (1 year).